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Olivier Amacker

Numerical Stability of Functional MRI Connectivity Biomarkers

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University of Applied Sciences and Arts Western Switzerland
Faculty of Engineering and Architecture
Haute École d'Ingénierie de Sion

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Olivier Amacker

Numerical Stability of Functional MRI Connectivity Biomarkers

Thesis submitted to the Bachelor degree programme
Informatique et Systèmes de communication (ISC) – Data engineering major
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Supervisor: **PROF. DR OSCAR ESTEBAN**
Co-supervisor: **DR OKITO YAMASHITA**

Expert: **DR AYUMU YAMASHITA**

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Abstract

As Functional Magnetic Resonance Imaging (fMRI) pipelines become more computationally intensive, requiring high-dimensional computations, it is crucial to assess how small numerical variability introduced by factors such as the running Operating System (OS), parallelization strategies, and hardware architecture affects the results of these pipelines.

Building upon the paper "Numerical Variability of Magnetic Resonance Imaging (MRI) Graph Measures" [1], this work extends the assessment beyond graph measures to examine Major Depressive Disorder (MDD) biomarkers, specifically the Principal Component Analysis (PCA) based biomarkers proposed by Yamashita et al., 2026 [2], by perturbing several steps of the pipeline. It also evaluates the impact of numerically perturbed fMRIPrep preprocessing on Functional Connectivity (FC) matrices using the Numerical Population Variability Ratio (NPVR) metric.

To reproduce the findings of the original study, the results were obtained by running the fMRIPrep [3] (25.2.5) preprocessing pipeline through a custom Docker container with floating-point arithmetic perturbations introduced by Verificarlo [4] and Fuzzy [5]. After preprocessing, FC matrices were obtained using Nilearn [6] (0.13.1). Different absolute thresholds were applied to the FC matrices, and graph metrics were computed using NetworkX [7] (3.6.1).

The graph metrics assessment, performed on a different multi-site dataset, showed consistent trends, though with higher NPVR values, and revealed that applying more confounds increases the NPVR as well as the variability of this effect across thresholds. The edge-wise FC analysis extended this work by quantifying numerical noise at the individual connection level and comparing the effect of including versus excluding the global signal confound. This comparison revealed substantial variability (mean NPVR of approximately 11%) that was notably reduced by half when global signal regression was applied.

For the assessment of the PCA feature selection, two steps of the pipeline were perturbed: the correlation coefficient computation using a specific Fuzzy Docker image, and the principal component analysis by forcing it to use 32-bit floating-point inputs rather than 64-bit.

The PCA feature extraction proved to be highly stable under the introduced numerical perturbations, yielding identical results across all perturbed runs on both the SRPB and BMB datasets. These findings suggest that the PCA features extraction is not only stable across different imaging sites and datasets, but also robust to numerical variability, supporting its reliability for clinical applications.

Keywords
medical data science, fMRI, fMRIPrep, numerical stability, bio-markers, PCA, neuroimaging, MDD

Résumé

Alors que les pipelines de Imagerie par Résonance Magnétique Fonctionnelle (IRMf) deviennent de plus en plus intensifs sur le plan computationnel, nécessitant des calculs en haute dimension, il est crucial d'évaluer comment de petites variations numériques introduites par des facteurs tels que le Système d'Exploitation (SE) utilisé, les stratégies de parallélisation et l'architecture matérielle affectent les résultats de ces pipelines.

S'appuyant sur le papier « Numerical Variability of IRMf Graph Measures » [1], ce travail étend l'évaluation au-delà des mesures de graphes pour examiner les biomarqueurs du Trouble Dépressif Majeur (TDM), notamment les biomarqueurs basés sur l'Analyse en Composantes Principales (ACP) proposés par Yamashita et al., 2026 [2], en perturbant plusieurs étapes du pipeline. Il évalue également l'impact du prétraitement fMRIPrep numériquement perturbé sur les matrices de Connectivité Fonctionnelle (CF) en utilisant la métrique Numerical Population Variability Ratio (NPVR).

Pour reproduire les résultats de l'étude originale, les résultats ont été obtenus en exécutant le pipeline de prétraitement fMRIPrep [3] (25.2.5) dans un conteneur Docker personnalisé avec des perturbations arithmétiques en virgule flottante introduites par Verificarlo [4] et Fuzzy [5]. Après prétraitement, les matrices de CF ont été obtenues à l'aide de Nilearn [6] (0.13.1). Différents seuils absolus ont été appliqués aux matrices de CF, et les métriques de graphes ont été calculées à l'aide de NetworkX [7] (3.6.1).

L'évaluation des métriques de graphes, réalisée sur un ensemble de données multi-sites différent, a montré des tendances cohérentes, bien qu'avec des valeurs NPVR plus élevées, et a révélé que l'application de plus de facteurs de confusion augmente le NPVR ainsi que la variabilité de cet effet à travers les seuils. L'analyse au niveau des connexions individuelles des matrices de CF a étendu ce travail en quantifiant le bruit numérique au niveau de chaque connexion et en comparant l'effet de l'inclusion versus l'exclusion du facteur de confusion du signal global. Cette comparaison a révélé une variabilité substantielle (NPVR moyen d'environ 11%) qui a été réduite de moitié lorsque la régression du signal global a été appliquée.

Pour l'évaluation de la sélection de caractéristiques par ACP, deux étapes du pipeline ont été perturbées : le calcul des coefficients de corrélation à l'aide d'une image Docker Fuzzy spécifique, et l'analyse en composantes principales en la forçant à utiliser des entrées en virgule flottante 32 bits plutôt que 64 bits.

L'extraction de caractéristiques par ACP s'est avérée très stable sous les perturbations numériques introduites, produisant des résultats strictement identiques pour toutes les exécutions perturbées sur les ensembles de données SRPB et BMB. Ces résultats suggèrent que l'extraction de caractéristiques par ACP est non seulement stable entre différents sites d'imagerie et jeux de données, mais également robuste face à la variabilité numérique, soutenant ainsi sa fiabilité pour les applications cliniques.

Mots-clés
medical data science, fMRI, fMRIPrep, numerical stability, bio-markers, PCA, neuroimaging, MDD

Declaration of Honour

Bachelor Thesis
Filière Informatique et systèmes de communication (ISC) · Haute École d'Ingénierie, HES-SO Valais-Wallis

Author :	Olivier Amacker
Thesis title :	Numerical Stability of Functional MRI Connectivity Biomarkers
Academic year :	2025-2026

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Chapter 1

Introduction

Functional Magnetic Resonance Imaging (fMRI) has revolutionized neuroscience by enabling non-invasive observation of brain activity through blood oxygenation level-dependent signals. A key application is the analysis of Functional Connectivity (FC), which examines temporal correlations between spatially distinct brain regions to construct connectivity matrices. These matrices serve as the foundation for deriving graph-theoretical metrics and biomarkers used in clinical research, particularly for conditions such as Major Depressive Disorder (MDD).

As fMRI pipelines become increasingly computationally intensive, involving high-dimensional matrix operations and complex preprocessing steps, the question of numerical reliability becomes crucial. Small numerical perturbations introduced by factors such as Operating System (OS) differences (as shown in Figure 1), hardware architecture, and parallelization strategies can propagate through the pipeline, potentially affecting downstream results.

Figure 1 - Example of the same program giving different results depending on the OS

Large Language Models (LLMs) were used during the writing of this report exclusively to reformulate sentences and correct typographical and grammatical errors. They were never used to generate text from scratch.

In the context of multi-site studies and clinical applications, where reproducibility is essential, understanding and quantifying this numerical variability is critical. Unstable biomarkers could lead to unreliable clinical decisions or failed replication across research sites. However, systematic assessment of numerical stability in fMRI pipelines remains limited, particularly for advanced feature extraction methods.

The goal of this bachelor thesis, conducted in collaboration with the Advanced Telecommunications Research Institute International (ATR), is to investigate the numerical stability of fMRI connectivity biomarkers. This work was divided into three complementary parts.

First, the objective was to reproduce the results of Alizadeh et al., 2026 [1], which examined how numerical variability affects the sample size required for statistical significance and the stability of FC matrix graph metrics using the Numerical Population Variability Ratio (NPVR) metric.

Second, the focus shifted to the numerical stability of the FC matrices themselves, rather than their derived graph metrics, providing edge-wise stability analysis.

Finally, the numerical stability of a feature extraction method based on Principal Component Analysis (PCA) was assessed, building on the work of Yamashita et al., 2026 [2]. This method extracts FC biomarkers robust to different sites and datasets. The stability assessment involved perturbing the correlation coefficient computation (`np.corrcoef`) and forcing the PCA to use 32-bit floating-point inputs rather than 64-bit.

Chapter 2

State of the Art

In recent years, different tools have been created to help the assessment of numerical variability. Notably Verificarlo Denis et al., 2018 [4] which is a tool used to compile programs with LLVM level arithmetic perturbations and Fuzzy [5] which consists of already perturbed Docker containers for certain Python libraries like Numpy or Scipy. More recently Fuzzy PyTorch Gonzalez-Pepe et al., 2026 [8] has been released, which extends this approach to deep learning workflows.

These tools rely on Monte Carlo Arithmetic (MCA) Parker et al., 1997 [9], a technique that introduces controlled random perturbations to floating-point operations during program execution. By repeatedly running the same pipeline with MCA instrumentation, researchers can quantify how numerical variability propagates through computational workflows.

Using these tools, the impact of numerical variability has been investigated across several neuroimaging domains such as functional, structural and diffusion imaging. As introduced in Section 1, Alizadeh et al., 2026 [1] evaluated the numerical variability of different graph measures derived from the FC matrices resulting from the widely used fMRIPrep Esteban et al., 2019 [3] preprocessing pipeline. Using the NPVR metric, they obtained values ranging from 0.1 to 0.2 for most graph metrics. These results were found to vary across brain regions, FC thresholds and confound regression strategies.

In structural imaging pipelines, Mirhakimi et al., 2025 [10] investigated numerical uncertainty in linear registration algorithms, demonstrating that small floating-point variations can lead to measurable differences in image alignment. Similarly, Chatelain et al., 2026 [11] quantified the impact of numerical variability on structural Magnetic Resonance Imaging (MRI) measures in Parkinson's disease, finding that numerical variation reached nearly one-third of population variability in multiple cortical and subcortical regions.

In diffusion imaging pipelines, Kiar et al., 2021 [12] showed that numerical uncertainty in analytical workflows leads to impactful variability in brain network measurements, revealing that different computational environments can produce substantially different connectivity results.

Interestingly, numerical variability has also been used as a resource rather than a limitation. Gonzalez-Pepe et al., 2026 [8] established that controlled numerical perturbations can serve as an effective data augmentation strategy for deep learning models, artificially expanding training datasets and improving model generalization.

However, these stability conclusions cannot be generalized across pipelines due to heterogeneous methodologies and implementations. While graph-theoretical metrics have been studied by Alizadeh et al., 2026 [1], edge-wise FC matrix stability remains unexplored. Additionally, the PCA-based feature extraction proposed by Yamashita et al., 2026 [2] has shown robustness across different sites and datasets, making its numerical stability verification crucial. This thesis addresses these gaps through edge-wise FC analysis and PCA-based extraction evaluation.

Chapter 3

Development and Methodology

This section describes the development process and methodology underlying this thesis. Beyond the thesis-specific code, this work included contributions to external repositories via Pull Requests (PRs), enhancements to Yamashita et al.'s PCA-based feature extraction package, and a correction to the NPVR simulation in Alizadeh et al. 2026's GitHub repository.

3.1 – Code availability

The code developed during this bachelor thesis is publicly available on GitHub. The repository is organized using git submodules to separate concerns: the `bachelor-thesis-report` submodule contains this Typst report, while the `bachelor-thesis-project` submodule contains all analysis code, including preprocessing pipelines, analysis notebooks, and job submission scripts for the ATR High Performance Computing (HPC) cluster.

An interactive project website is available at olivier.amacker.dev/bachelor-thesis, providing browsable access to the four analysis notebooks developed with Marimo and their respective outputs, a daily journal documenting the project's progress, and the different presentations delivered during the thesis.

The development environment is managed via Nix and uv, ensuring its reproducibility. The fuzzy fMRIPrep Docker container, built on Verificarlo and the Fuzzy's libmath library, used for running perturbed preprocessing runs, is published on DockerHub for reproducibility.

3.2 – Reproduction of Alizadeh et al., 2026

This section details the reproduction of the two distinct analyses presented in Alizadeh et al., 2026 [1]. The original code can be found on GitHub, and before starting the reproduction, a comprehensive summary of the paper was created to ensure a thorough understanding of the methodology, preprocessing pipeline, and graph metrics which is accessible here. The first analysis involved implementing NPVR calculation on simulated data to assess how NPVR variation influences the sample size required for Cohen's d coefficient. The second analysis used perturbed fMRIPrep outputs to evaluate NPVR variation across different graph metrics, thresholds, and brain regions.

3.2.1 – NPVR Simulation

An interactive notebook for this simulation can be accessed online, as it doesn't rely on any external data.

The simulation begins by generating two synthetic populations using normal distributions with the same means but different standard deviations: $\sigma = 0.1$ for the low variability group and $\sigma = 0.4$ for the high variability group. For each population, the NPVR is calculated using the formula $\mathrm{NPVR} = \frac{\sigma_{\mathrm{num}}}{\sigma_{\mathrm{pop}}}$, where σ_{num} represents the numerical variability and σ_{pop} the population variability. The populations are then visualized in the σ_{num} / σ_{pop} space to assess their relative positions on NPVR contour lines. Finally, the relationship between NPVR, sample size, and Cohen's d variations is visualized to assess how numerical noise propagates into statistical inference.

During this reproduction, the original source code was reviewed and an error was identified. The NPVR values in the final visualization were hardcoded to `0.287` and `0.496` rather than computed from the generated distributions. This bug was fixed through a PR that was merged into the original repository.

3.2.2 – Graph Metrics Stability Assessment

The notebook code developed for this step and its outputs are available as a static HTML page.

This reproduction step involved several changes to the original procedure. First, a custom Docker container was created to perturb the latest version of fMRIPrep (25.2.5) instead of using the existing Fuzzy container, which used an older fMRIPrep version (23.2.1). Second, the fMRI data used for this section was provided by the ATR and originates from a multi-site dataset Koike et al., 2021 [13], rather than the publicly available PPMI dataset that was originally used. Third, when extracting the FC matrices, the original paper used six confound regressors corresponding to the main motion parameters (translations and rotations). However, as the data used in this work appeared to be noisier, nine additional confounds were included following the recommendation of Dr. Yamashita: global signal, CSF, white matter, and six aCompCor components (`a_comp_cor_00` through `a_comp_cor_05`). Fourth, due to computational constraints, the small-worldness graph metric was excluded from the analysis. Finally, how the NPVR is calculated based on the outputs of the different fMRIPrep runs was modified. In the original paper, they had the same number of runs per subject and the NPVR was calculated as follows:

$$\sigma_{\text{num}}^2 = \frac{1}{m}\sum_{j=1}^{m}\left[\frac{1}{n-1}\sum_{i=1}^{n}\left(x_{i,j} - x - |_{.,\mathrm{j}}\right)^2\right]$$

$$\sigma_{\text{pop}}^2 = \frac{1}{n}\sum_{i=1}^{n}\left[\frac{1}{m-1}\sum_{j=1}^{m}\left(x_{i,j} - x - |_{i,.}\right)^2\right]$$

$$\text{NPVR} = \frac{\sigma_{\text{num}}}{\sigma_{\text{pop}}}$$

where n is the number of MCA repetitions, m is the number of subjects, and $x_{i,j}$ is the metric value for subject j in MCA run i.

Initially, some preprocessing runs failed, resulting in varying numbers of MCA runs per subject. However, after rerunning the failed subjects, the final dataset contained 5 subjects with 5 fuzzy fMRIPrep runs each. Although the final dataset was balanced, the pooled approach was retained to handle potential missing data robustly. The pooled numerical variance accounts for the varying number of runs per subject:

$$\sigma_{\text{num pooled}}^2(r) = \frac{1}{\sum_{j=1}^{m}(n_j - 1)}\sum_{j=1}^{m}(n_j - 1)\left[\frac{1}{n_j - 1}\sum_{i\in\Omega_j}\left(x_{i,j}(r) - x - |_{.,\mathrm{j}}(r)\right)^2\right]$$

Similarly, the pooled population variance accounts for the varying number of subjects per run:

$$\sigma_{\text{pop pooled}}^2(r) = \frac{1}{\sum_{i=1}^{n}(m_i - 1)}\sum_{i=1}^{n}(m_i - 1)\left[\frac{1}{m_i - 1}\sum_{j\in\Omega_i}\left(x_{i,j}(r) - x - |_{i,.}(r)\right)^2\right]$$

The pooled NPVR for each region r is then:

$$\text{NPVR}_{\text{pooled}(r)} = \frac{\sigma_{\text{num pooled}(r)}}{\sigma_{\text{pop pooled}(r)}}$$

where n_j is the number of runs for subject j, m_i is the number of subjects in MCA run i, and Ω_j and Ω_i denote the sets of valid indices for subject j and run i, respectively.

This approach ensures that all available data contributes to the variability estimates, even if some runs are missing.

The remaining analysis followed the same procedure as the original study. After fMRIPrep preprocessing, the Schaefer 2018 parcellation [14] with 100 cortical regions and 7 functional networks was used to extract regional time series using Nilearn's NiftiLabelsMasker. Spatial smoothing of 6mm Full Width at Half Maximum (FWHM), temporal standardization, and detrending were applied during masking. The FC matrices were computed as Pearson correlation matrices using Nilearn's ConnectivityMeasure, with both with-confound and without-confound versions generated. The matrices were thresholded using absolute correlation values at six thresholds: 0.05, 0.1, 0.2, 0.3, 0.4, and 0.5, retaining both strongly positive and strongly negative correlations. For each threshold, four local graph metrics were computed using NetworkX: degree centrality, clustering coefficient, betweenness centrality, and eigenvector centrality. Additionally, one global metric was calculated: average shortest path length. The NPVR was then computed for each metric and threshold. For visualization purposes, these values were normalized and plotted across brain regions by projecting regional values onto the Schaefer 2018 atlas to assess spatial variability in numerical stability. Furthermore, the NPVR was computed on the difference between with-confound and without-confound matrices to assess the impact of confound regression, following the approach of the original study.

3.3 – Edge-wise FC Matrix Stability Analysis

The notebook code developed for this step and its outputs are available as a static HTML page.

Reusing the perturbed fMRIPrep runs from Section 3.2.2, this section extends the analysis from graph-level metrics to individual edges of the FC matrices. The pooled NPVR was computed independently for each of the 4950 edges (upper triangle of the 100x100 matrix) using the same formula as before.

Two confound strategies were compared: the full set of 15 confounds and a filtered set excluding only the global signal. The global signal regression remains a debated practice in the neuroimaging community, as it may remove both nuisance signals and neural activity of interest Xu et al., 2018 [15], Xifra-Porxas et al., 2025 [16]. To quantify its impact on edge-wise numerical stability, a delta NPVR was computed for each edge as the difference between the two confound strategies.

The edge-wise NPVR values were visualized as connectivity matrices, scatter plots, and histograms. To assess spatial patterns, edge-wise values were aggregated per brain region by computing the mean and median NPVR across all connected edges and mapped onto the Schaefer 2018 atlas.

3.4 – PCA-based Feature Extraction Stability

A browsable version of the analysis notebook is available online.

This section had three objectives. First, reproduce the PCA-based feature extraction method developed by Yamashita et al., 2026 [2] on the complete SRPB dataset [17], as opposed to the original paper which split the data into discovery and validation sets. Second, apply the same method to the complete BMB dataset [13]. Third, perturb different steps of the pipeline: the correlation computation using a MCA-instrumented Docker container and the PCA dimensionality reduction by forcing 32-bit floating-point inputs instead of the standard 64-bit precision.

3.4.1 – Reproduction on the SRPB and BMB Datasets

The original PCA feature extraction code was publicly available on GitHub but organized as separate Python files, making integration challenging. A PR was submitted and merged to restructure it into the `pcafeat` package with uv as a dependency manager.

The FC matrices used as inputs to the pipeline had already been preprocessed and harmonized across imaging sites by ATR for both datasets using the method of Yamashita et al., 2019 [18], so they were used directly without any additional preprocessing or harmonization step.

The extraction pipeline was applied to six metrics: diagnosis (MDD vs. HC), BDI score, age, sex, imaging site, and mean Framewise Displacement (FD). The pipeline works as follows:

1. A PCA reduces all n subjects' FC vectors to n Principal Components (PCs)
2. Each PC is tested for association with the desired target variables using appropriate statistical tests, with results plotted during testing:
   - t-tests for binary categories (e.g., sex)
   - ANOVA for multi-level categories (e.g., site)
   - Pearson's correlation for continuous variables (e.g., age)
3. PCs significantly associated with the target after False Discovery Rate with Benjamini-Hochberg (FDR-BH) correction ($q < 0.05$) are selected
4. Their FC weights are tested via a chi-squared test with the same correction threshold
5. Only FCs surviving the corrections are retained as final biomarkers

During replication of the original paper's pipeline, the plots generated during PC association testing were found to have missing labels and incomplete titles. This issue was fixed in another PR.

After running the extraction pipeline on both datasets, the selected FCs for the second PC were plotted on brain surface maps using a combined parcellation of 446 Regions Of Interest (ROIs): 360 cortical regions from the Glasser et al., 2016 [19] atlas, 54 subcortical regions from the Tian et al., 2020 [20] atlas, and 32 cerebellar regions from the Nettekoven et al., 2024 [21] atlas. Functional network assignments were derived from the Yeo et al., 2011 [22] 7-network parcellation. Each selected FC connection was visualized as an edge between brain regions, color-coded by direction of effect: red indicating over-connectivity in MDD relative to healthy controls, and blue indicating under-connectivity. Nodes were colored according to their functional network assignment (Visual, SomatoMotor, DorsalAttention, DefaultMode, Limbic, Salience, PrefrontalControl, Subcortical, or Cerebellum).

Network statistics were also computed to quantify the distribution of selected connections across brain networks, including the number of participating nodes per network, the proportion of intra-network versus inter-network edges, and the most frequent inter-network connection pairs.

3.4.2 – FC Matrix Regular Extraction and Harmonization Pipeline

While Section 3.4.1 used FC matrices already preprocessed and harmonized, the perturbation strategy described in the next section targets the correlation coefficient computation (`np.corrcoef`) used during the FC matrices extraction. Reproducing the extraction and harmonization pipeline from the raw parcellated time series was therefore necessary to establish a numerically consistent pipeline before introducing any perturbation.

The starting point was the preprocessed regional time series provided by ATR for the SRPB dataset. These time series were parcellated according to the previously described combined parcellation (360 cortical, 54 subcortical, and 32 cerebellar regions, 446 ROIs in total), with global signal regression and band-pass filtering already applied.

FC extraction proceeded as follows. First, the regional time series were scrubbed for motion by removing any frame with FD exceeding 0.5 mm, together with the immediately subsequent frame. On the retained frames, Pearson correlation matrices were computed using `np.corrcoef` applied to the transposed time series array. The lower triangular elements of each correlation matrix were extracted, Fisher Z-transformed via `np.arctanh`, and flattened into a subject-level connectivity vector.

Before harmonization, the self-extracted FC matrices were validated against the original ATR-provided harmonized matrices. For each subject, the Pearson correlation and mean absolute error between the extracted and reference connectivity vectors were computed. The resulting per-subject correlation distribution showed near-perfect agreement (mean: 0.9970, median: 0.9967, min: 0.9852), confirming that the extraction pipeline faithfully reproduced the original connectivity values prior to harmonization (Figure 2).

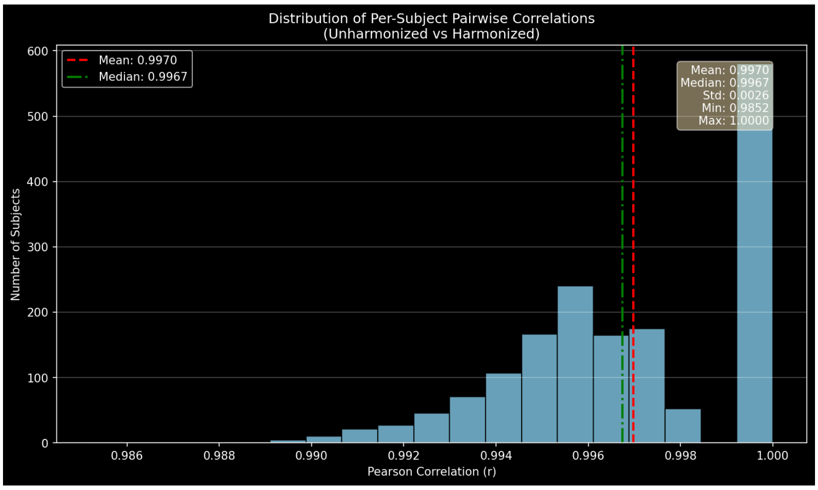


Figure 2 - Per-subject Pearson correlation distribution between self-extracted and ATR-provided FC connectivity vectors before harmonization.

Cross-site harmonization was then performed using the traveling-subject method of Yamashita et al., 2019 [18]. This approach models unwanted variance as a linear combination of subject, site, sampling, and protocol biases. Dummy variables were created for each categorical factor, and the sampling dummies were orthogonalized against the site dummies using a weighted projection to ensure independence between site and sampling effects. The bias estimation was formulated as a constrained regularized least-squares problem. In the corresponding Karush–Kuhn–Tucker (KKT) optimality conditions, the Hessian matrix encodes the second-order curvature of the regularized objective function. Rather than inverting this large matrix directly, the system was solved efficiently by first factorizing the Hessian via Cholesky decomposition, then applying a Schur-complement reduction to eliminate the primal variables and obtain a much smaller linear system for the Lagrange multipliers. Generalized cross-validation was used to select the regularization hyperparameter λ.

Once the bias coefficients were estimated, the relevant site and protocol bias terms were subtracted from each regular subject's connectivity vector, yielding harmonized FC matrices. The self-harmonized matrices were again compared against the ATR-provided harmonized matrices. The near-zero mean difference and high per-subject correlation (mean: 0.9999, median: 0.9999, min: 0.9996) confirmed that the entire reproduction pipeline from scrubbed time series to harmonized connectivity was numerically consistent with the original processing, thereby establishing a valid baseline for the subsequent perturbation experiments (Figure 3). The identical procedure was subsequently applied to the BMB dataset.

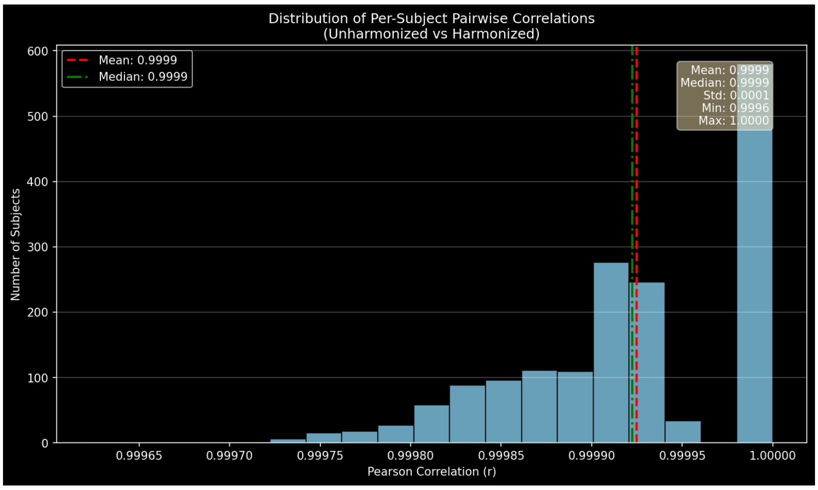


Figure 3 - Per-subject Pearson correlation distribution between self-harmonized and ATR-provided harmonized FC connectivity vectors.

3.4.3 – FC Matrix Perturbed Extraction

Once the regular pipeline was validated, a perturbed version was implemented by executing `np.corrcoef` inside an MCA-instrumented Docker container provided by Fuzzy. Within this container, Verificarlo introduces controlled random perturbations to each floating-point operation during correlation computation, simulating numerical noise arising from hardware and OS variability. One hundred perturbed extractions were performed on the SRPB dataset ($n = 1667$), while fifteen perturbed extractions were performed on the provided subset of the BMB dataset ($n = 6371$). The lower number of runs for the BMB dataset reflects its larger subject count and corresponding computational cost. Each run yielded a complete set of subject-level connectivity vectors used for numerical stability comparison.

3.4.4 – PCA Dimensionality Reduction Perturbation

Additionally, the PCA feature extraction step was also perturbed by using the already MCA-perturbed FC matrices as inputs, cast to single precision (32 bits) instead of the default double precision (64 bits). This reduces the representable precision from approximately 16 to 7 significant decimal digits. The numerical stability assessment therefore reflects the combined effect of both perturbation sources on the same data.

Chapter 4

Results and Discussion

This section showcases the results of the three complementary analyses described in Section 3 and discusses their implications. First, the reproduction of Alizadeh et al., 2026 [1] graph metrics assessment evaluates how numerical variability affects downstream graph measures. Second, the edge-wise FC analysis quantifies numerical noise at the level of individual correlations before any graph construction. Third, the stability of the PCA-based feature extraction pipeline is assessed under perturbations of both correlation computation and precision reduction.

4.1 – Reproduction of Alizadeh et al., 2026

This part reproduced both analyses of Alizadeh et al., 2026 [1]. The NPVR simulation on synthetic data (Section 3.2.1) was reimplemented to verify how numerical variability propagates into statistical inference. The graph metrics assessment (Section 3.2.2) was then rerun on fMRIPrep-preprocessed data from a different multi-site dataset, in order to test whether the original numerical stability findings generalize to other data sources.

4.1.1 – NPVR Simulation

Because the NPVR simulation relies solely on randomly generated synthetic data and the different factors except the small amount of randomness were preserved, the results were expected to closely match those of the original paper. This is examined for each of the three visualisations below.

First, the synthetic populations generated in the reproduction closely resemble those presented in the original paper. This similarity is clearly illustrated by Figure 4.

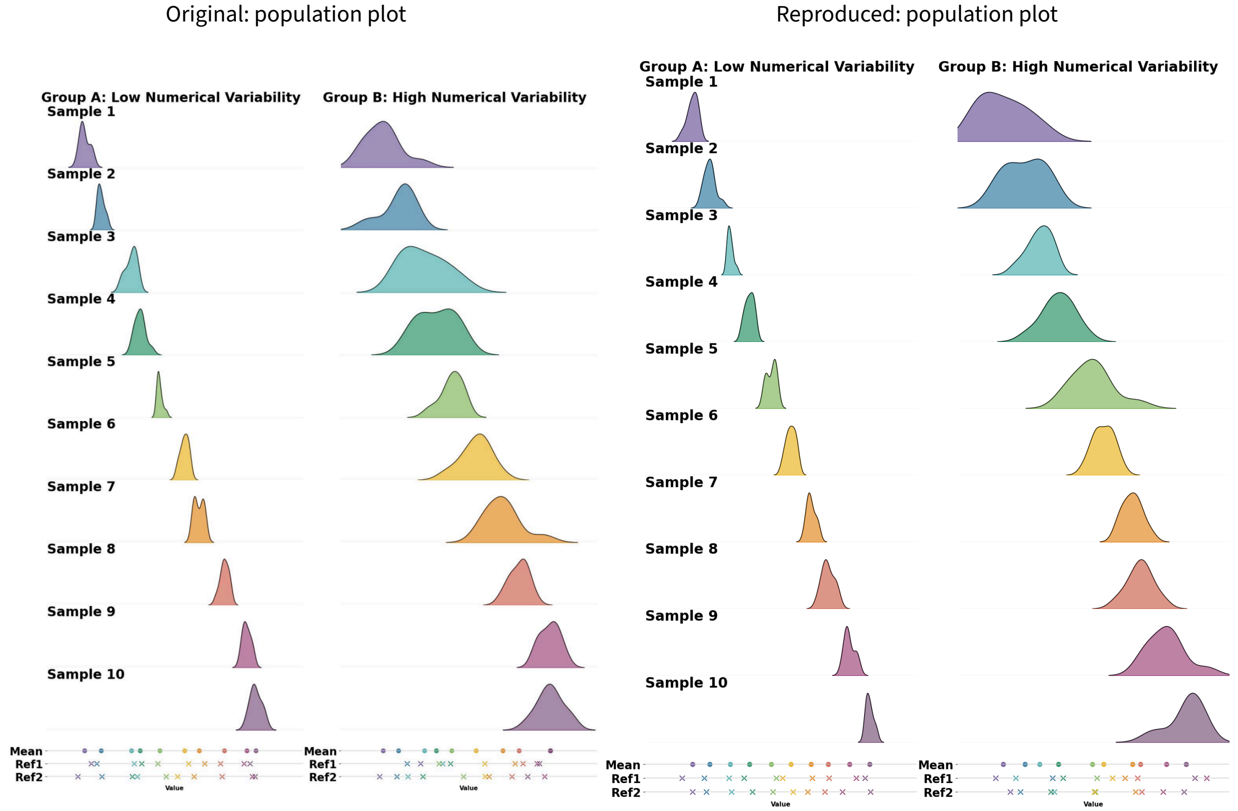


Figure 4 - Comparison of the original and reproduced synthetic populations.

Although the exact population values differ because `np.random.normal` produces independent random draws, the low and high variability groups display the same distributional characteristics as in the original analysis. This confirms that the population generation step is correctly implemented and reproducible.

Second, the NPVR of both the low and the high variability populations was calculated and plotted relatively to how the NPVR changes depending on the numerical and population variability.

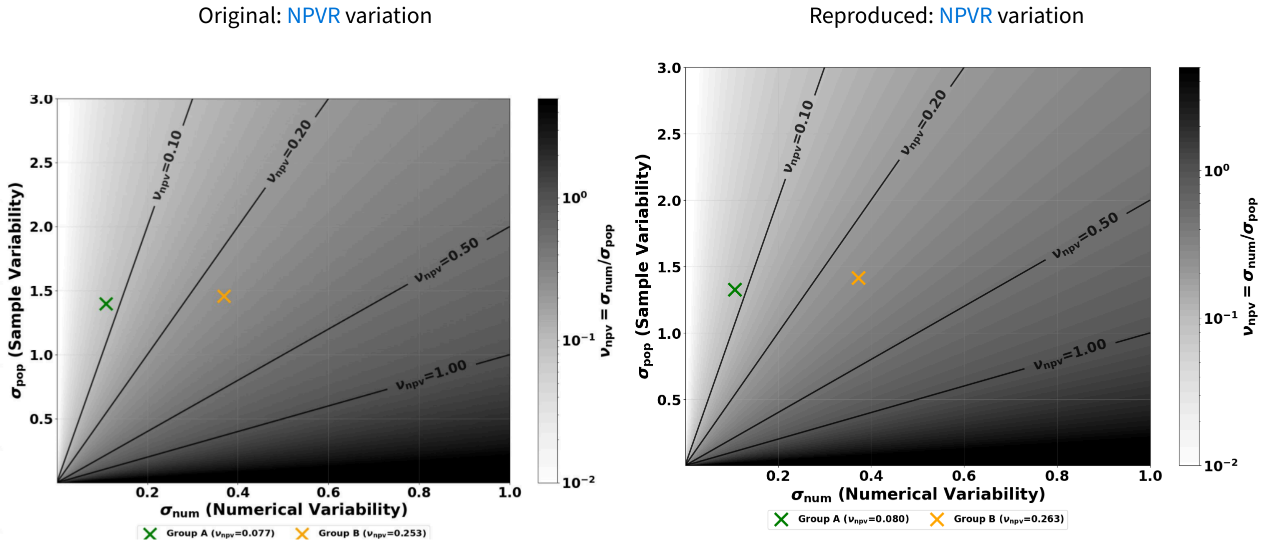


Figure 5 - Comparison of the original and reproduced NPVR variation plots.

In Figure 5, the original and reproduced NPVRs are in close agreement: 0.077 versus 0.080 for the low-variability population and 0.253 versus 0.263 for the high-variability population. This confirms that, despite minor random differences in the initial populations, the NPVR estimations are robust and the reproduction faithfully recovers the original values.

Third, the relationship between NPVR, sample size, and Cohen's d was also visualized. However, unlike the previous two plots, the original and reproduced figures are not expected to match. As detailed in Section 3.2.1, the original code hardcoded the NPVR values to 0.287 and 0.496 rather than computing them from the generated distributions. The reproduced plot corrects this by using the actual NPVR values calculated from the synthetic populations. Because no corrected version of the original figure has been released since the PR fixing this issue was merged, the original figure is shown here for reference only.

Original: hardcoded NPVR Cohen's d effect

Reproduced: computed NPVR Cohen's d effect

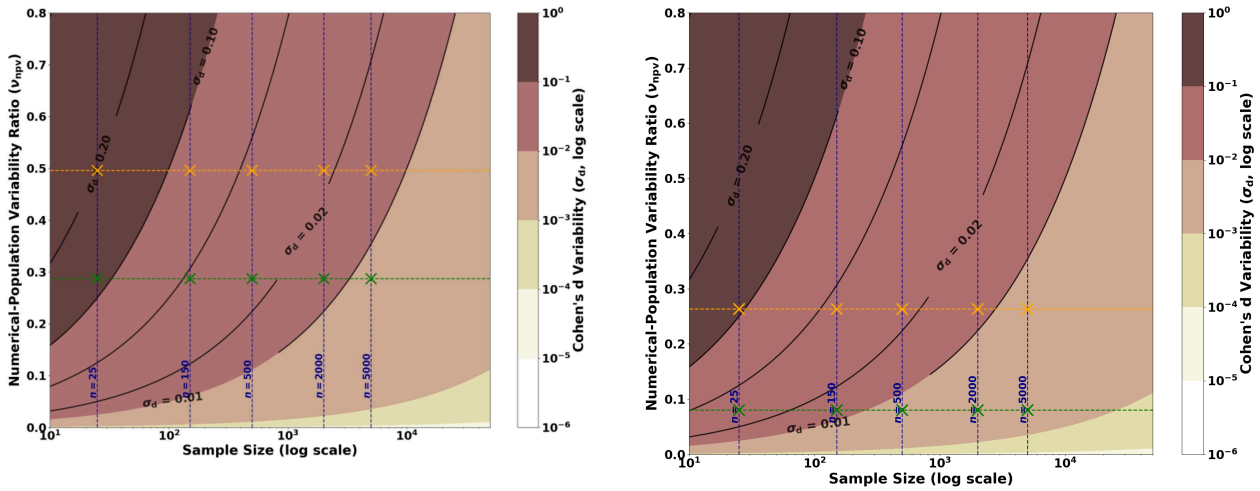


Figure 6 - Comparison of the original and reproduced effect of the NPVR on Cohen's d plots.

Clearly visible in Figure 6, the differences between the two panels confirm that the original published figure was affected. The reproduced plot correctly reflects the computed relationship between the NPVR, sample size, and Cohen's d.

4.1.2 – Graph Metrics Stability Assessment

The reproduction of the graph metrics assessment differed from the original study in two notable ways, as detailed in Section 3.2.2. First, a different multi-site dataset was used as input for the perturbed fMRIPrep runs. Second, a more extensive confound regression strategy was applied (15 confounds versus the original 6). These methodological differences make direct quantitative comparison of the NPVR values difficult, as any observed difference could originate from the data characteristics or the confound strategy rather than from the reproduction itself. Nevertheless, across the five computed graph metrics, the reproduced NPVR values computed by aggregating all runs, including both with-confound and without-confound matrices showed trends consistent with those reported by the original paper.

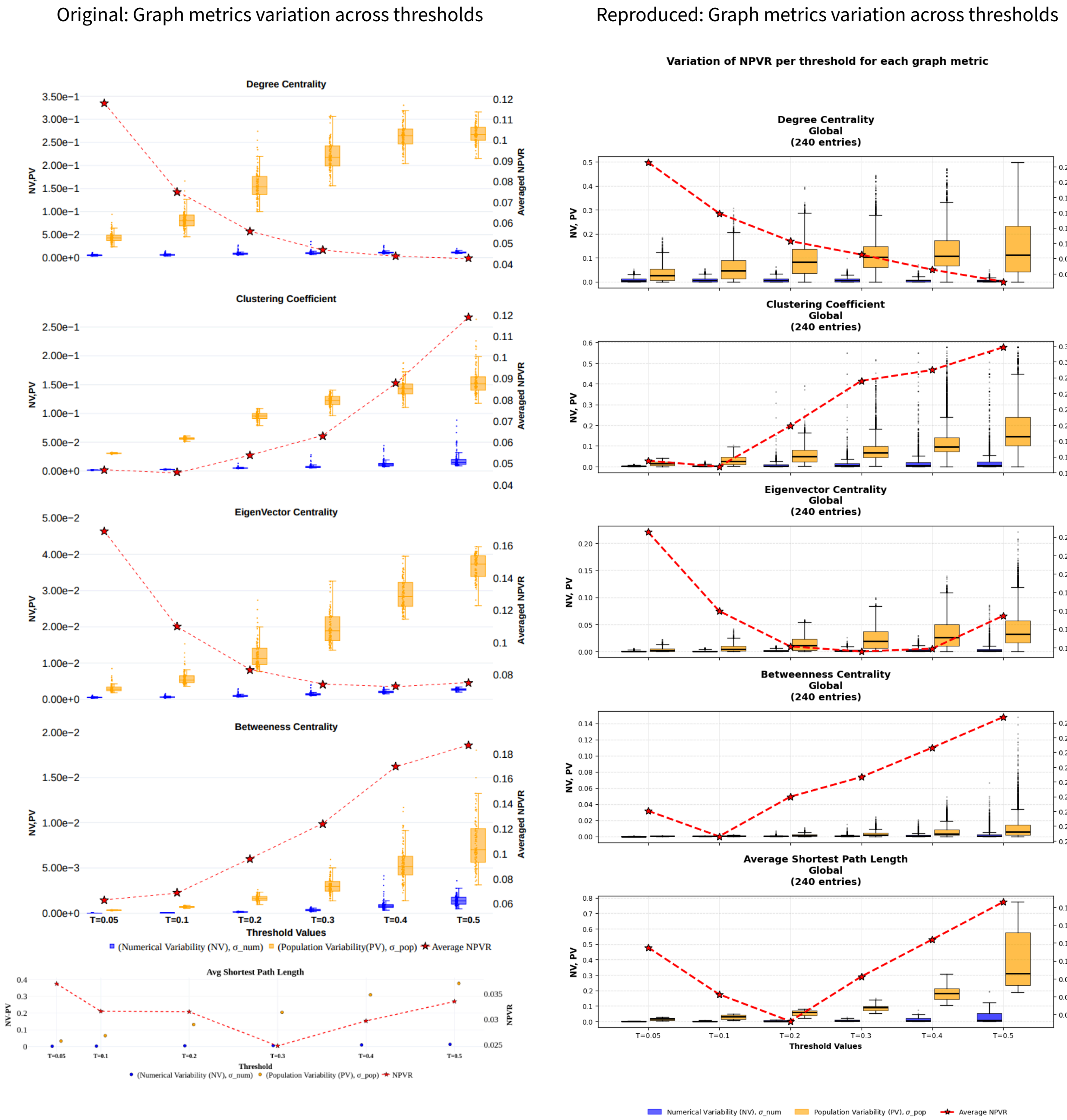


Figure 7 - Comparison of the original and reproduced NPVR values across graph metrics and thresholds

According to Figure 7, although the reproduced graph metrics display broadly similar trends, notable differences remain. First, eigenvector centrality increases at the 0.5 threshold in the reproduced data, whereas it remains at a lower level in the original results. Second, for the average shortest path length, the reproduced NPVR at the 0.5 threshold appears higher than at the 0.1 threshold, a pattern not observed in the original analysis. Finally, despite the application of a more extensive confound regression strategy, the reproduced NPVR values are consistently higher than those originally reported across the different thresholds and graph metrics.

Given the consistently higher NPVR values observed in the reproduction, the effect of confound regression strategies was expected to differ from the original findings. In the original paper, applying the six standard translation and rotation confounds consistently increased the NPVR for most graph metrics, though betweenness centrality and average shortest path length showed decreased values at the 0.1 threshold.

Original: Confound regression effect on NPVR

Reproduced: Confound regression effect on NPVR

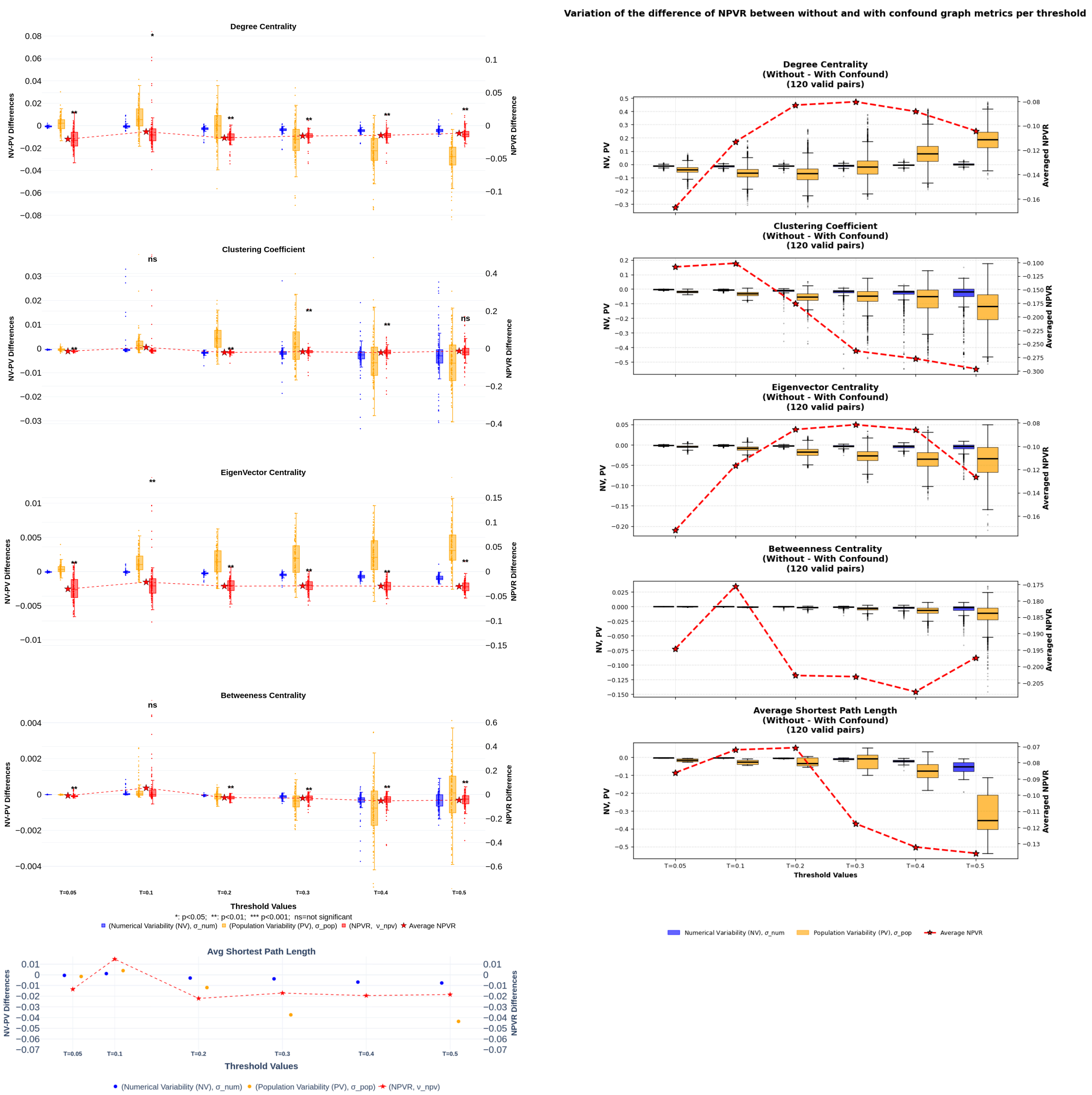


Figure 8 - Comparison of the NPVR difference between without-confound and with-confound strategies across graph metrics and thresholds.

As Figure 8 demonstrates, the reproduced confound regression effects differ substantially from those reported in the original paper. Unlike the original analysis, the reproduced data exhibits a consistently negative difference across all metrics and thresholds, indicating that confound regression uniformly increases the NPVR without exception. Moreover, both the magnitude and the variation of this effect are considerably larger in the reproduced data, with differences spanning a much wider range, particularly for clustering coefficient and average shortest path length at higher thresholds. This amplification is likely attributable to the use of more confound regressors rather than to the different dataset, as the effect follows the same direction as the original study, suggesting that the more confounds applied, the greater the increase in NPVR.

Finally, the NPVR was also evaluated across brain regions and found to vary spatially in both the reproduced and original data. For visualization purposes, these values were normalized per metric to enable comparison across different scales. However, Figure 9 reveals that the substantial differences in NPVR values between the two analyses result in distinct spatial patterns, making direct visual comparison of the brain surface plots difficult.

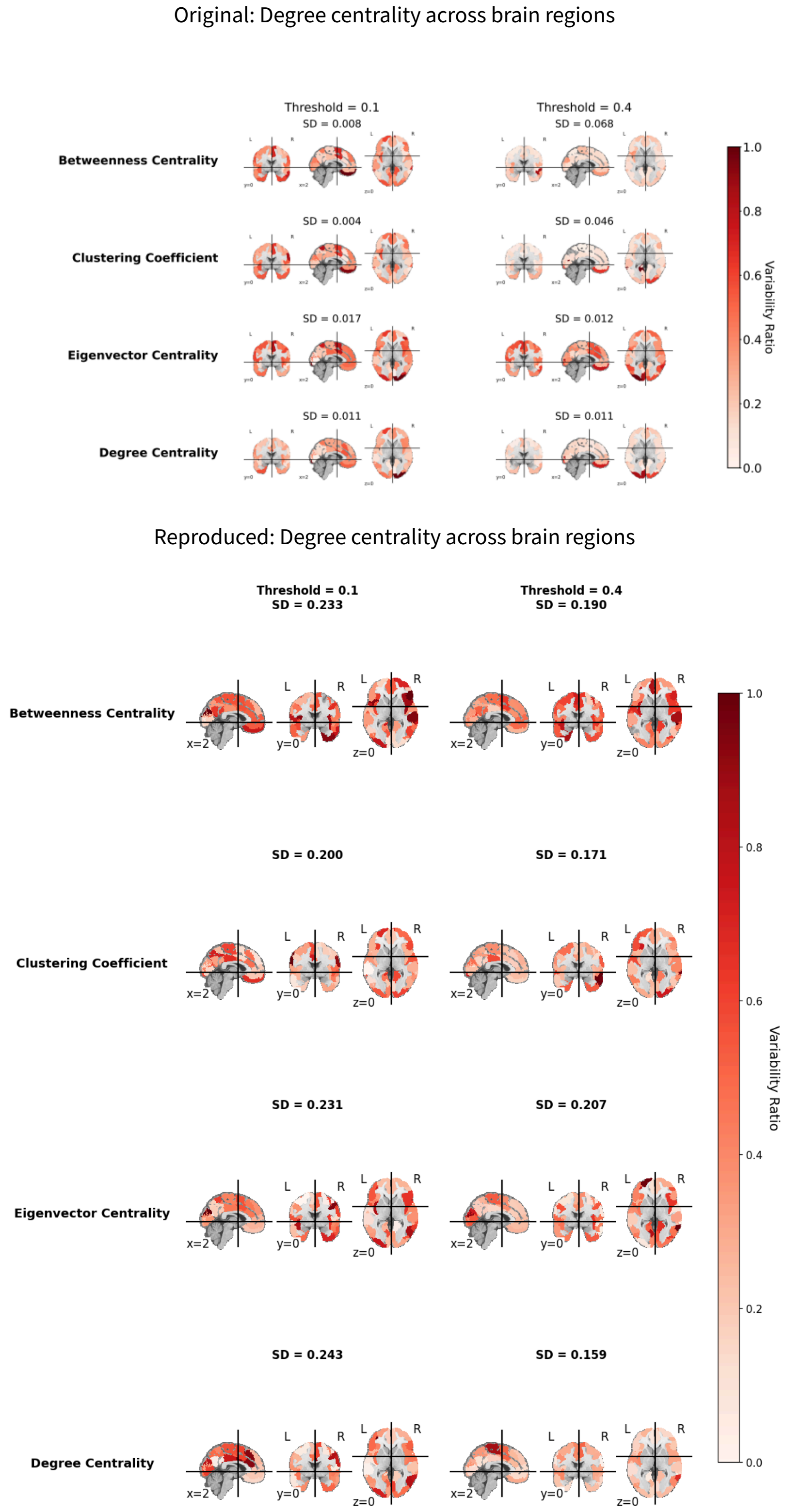


Figure 9 - Comparison of the NPVR across the different brain regions and thresholds for the degree centrality metric.

The reproduction of Alizadeh et al., 2026 [1] produced both confirming and diverging results. The NPVR simulation on synthetic data was faithfully reproduced, yielding results closely matching the original and enabling the identification and correction of a bug in the original code. The graph metrics assessment showed broadly consistent trends across thresholds and metrics, though the reproduced NPVR values were consistently higher than those originally reported. The confound regression analysis revealed that the additional regressors applied in the reproduction uniformly increased the NPVR across all metrics and thresholds. This is consistent with the original findings, which showed that even the six standard confounds increased the NPVR for most metrics, suggesting that the more confounds applied, the greater the increase in NPVR. Finally, while spatial variability of NPVR across brain regions was observed in both analyses, the substantial differences in NPVR values prevented direct visual comparison of the brain surface plots. Overall, these results suggest that while the general patterns of numerical variability are reproducible, the specific NPVR values are sensitive to dataset characteristics and preprocessing choices, particularly the confound regression strategy.

4.2 – Edge-wise FC Matrix Stability Analysis

As described in Section 3.3, the perturbed fMRIPrep outputs were reused to assess how the FC matrices themselves were affected by numerical perturbations, rather than their derived graph metrics.

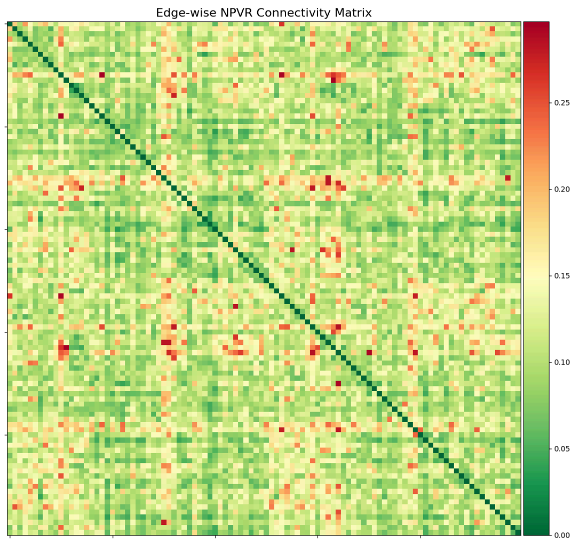


Figure 10 - Heatmap of NPVR values across the edges of the FC matrix.

The heatmap in Figure 10 confirms the spatial variability observed in Figure 9, illustrating that the NPVR is not uniformly distributed across the FC matrix, ranging from 0.0364 to 0.2970. Based on Figure 11, the mean and median values of 0.112 and 0.118 respectively represent a substantial amount of numerical variability, corresponding to approximately 11% of the population variability across edges.

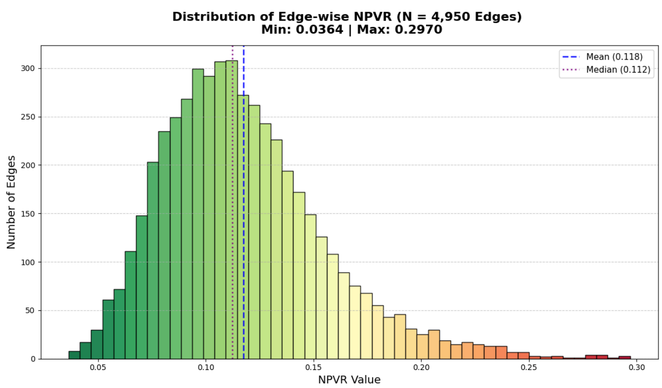
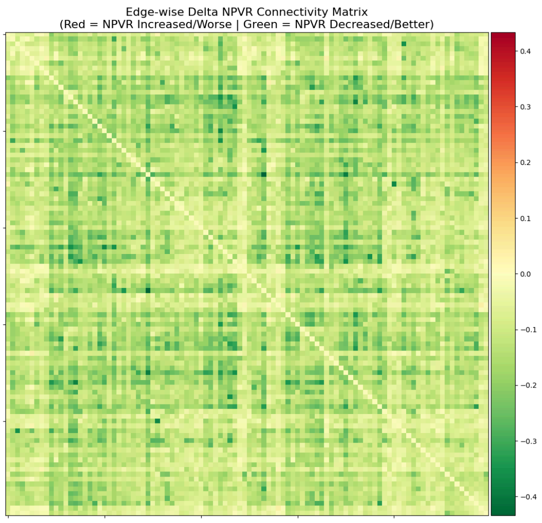


Figure 11 - Histogram of NPVR values across the edges of the FC matrix.

We assumed in Section 4.1.2 that using more confounds would increase the NPVR. However, by looking at the effect of the specific global-signal confound, the data demonstrate that this is not the case for all confounds. In fact, this specific confound appears to reduce the NPVR of the whole FC matrix by more than half, as quantified in Figure 12, with a mean and median delta of -0.126 and -0.119 respectively, and values ranging from $+0.0362$ to -0.4337.

Delta NPVR heatmap



Delta NPVR histogram

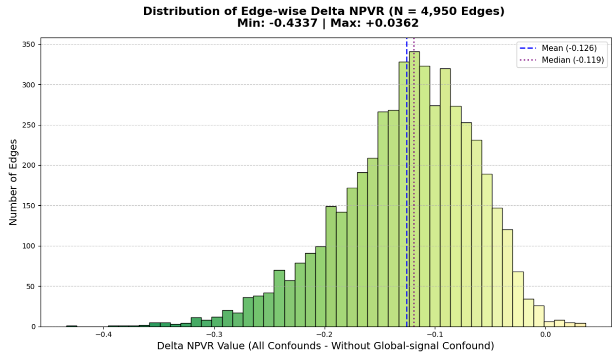


Figure 12 - Effect of global signal regression on the NPVR values across the edges of the FC matrix.

The heatmap in Figure 12 also demonstrates that the NPVR reduction is widespread across the entire FC matrix rather than concentrated in specific regions, with only a small fraction of edges showing increased variability.

The edge-wise analysis revealed substantial numerical variability across the FC matrix, with mean and median NPVR values of approximately 11% of the population variability. The global signal regression confound was found to significantly reduce this variability, halving the NPVR across the vast majority of edges. However, this analysis only examined the effect of a single confound. A more comprehensive assessment of how each individual confound contributes to numerical stability would provide valuable insights into optimal preprocessing strategies and could be a promising direction for further analysis.

4.3 – PCA-based Feature Extraction Stability

This section presents the results of the PCA-based feature extraction pipeline. First, the extraction is applied to both the SRPB and BMB datasets and compared against the findings of Yamashita et al., 2026 [2]. The combined impact of numerical noise introduced during FC matrix extraction and reduced input precision for the PCA on the extracted features is then assessed.

4.3.1 – Reproduction on the SRPB and BMB Datasets

The results obtained by the original paper already appeared reasonably consistent since they used a splitting of the SRPB dataset into a discovery and validation set, and obtained similar results. However, it is still important to see if these results stay similar when taking the entire dataset, or even a different one. The comparisons below focus on the second principal component (PC 2), as it consistently emerged as the most informative component across all analyses.

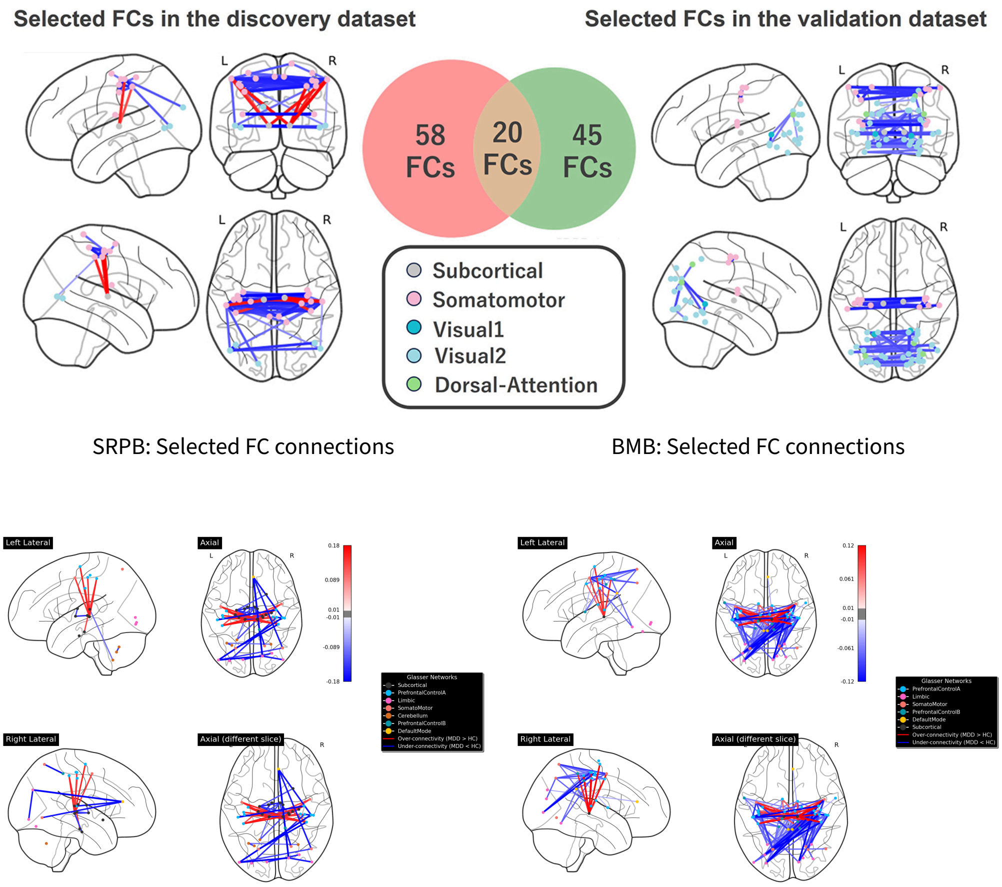


Figure 13 - Selected FC connections for PC 2 from the original paper (top) and from the SRPB and BMB datasets (bottom). Red edges indicate over-connectivity in MDD relative to healthy controls, blue edges indicate under-connectivity.

Figure 13 compares the original results (top) with those obtained on the SRPB and BMB datasets (bottom). Red edges indicate over-connectivity in MDD relative to healthy controls, blue edges indicate under-connectivity. The specific FC connections selected by the PCA feature extraction differ across all comparisons, whether considering the discovery and validation splits of the original dataset, the complete SRPB and BMB datasets, or each individual dataset against the original splits. However, analyzing the direction of the connectivity differences reveals a consistent pattern: the vast majority of connections show under-connectivity in MDD subjects across all datasets. The distribution of these connections across functional networks for both full datasets is detailed in Figure 14. While the original paper highlighted thalamic and motor regions as the primary drivers, our analysis reveals a broader and more variable involvement of prefrontal, motor, and subcortical networks. Although the relative dominance of these networks varies between the SRPB and BMB datasets, their consistent presence across analyses suggests they are key drivers of the identified connectivity patterns.

SRPB: Edges per network

FC Edges per Network

Network	Total Edges	Intra-Network	% of Total
Subcortical	62	18	52.5%
PrefrontalControlA	28	2	23.7%
Limbic	10	3	8.5%
SomatoMotor	8	0	6.8%
Cerebellum	6	2	5.1%
DefaultMode	3	0	2.5%
PrefrontalControlB	1	0	0.8%

SRPB: Inter-connections

Top Inter-Network Connections

Connection Pair	Count
PrefrontalControlA ↔ Subcortical	20
SomatoMotor ↔ Subcortical	4
Limbic ↔ PrefrontalControlA	2
Cerebellum ↔ Subcortical	2
PrefrontalControlB ↔ SomatoMotor	1
DefaultMode ↔ Limbic	1
DefaultMode ↔ PrefrontalControlA	1
DefaultMode ↔ SomatoMotor	1
Limbic ↔ SomatoMotor	1
PrefrontalControlA ↔ SomatoMotor	1

BMB: Edges per network

FC Edges per Network

Network	Total Edges	Intra-Network	% of Total
PrefrontalControlA	128	11	42.1%
SomatoMotor	66	8	21.7%
PrefrontalControlB	38	3	12.5%
Limbic	33	0	10.9%
Subcortical	28	0	9.2%
DefaultMode	11	0	3.6%

BMB: Inter-connections

Top Inter-Network Connections

Connection Pair	Count
PrefrontalControlA ↔ SomatoMotor	32
Limbic ↔ PrefrontalControlA	26
PrefrontalControlA ↔ Subcortical	22
PrefrontalControlA ↔ PrefrontalControlB	19
PrefrontalControlB ↔ SomatoMotor	8
DefaultMode ↔ PrefrontalControlA	7
SomatoMotor ↔ Subcortical	4
DefaultMode ↔ SomatoMotor	3
Limbic ↔ SomatoMotor	3
Limbic ↔ PrefrontalControlB	3

Figure 14 - Network statistics for SRPB and BMB datasets, showing edges per network and inter-network connections.

4.3.2 – Results of the perturbed extraction

Comparing the results across the different perturbed runs allows us to assess whether the PCA feature extraction is stable not only across different multi-site datasets, but also under numerical perturbations. We first examine the impact of the `np.corrcoef` perturbations detailed in Section 3.4.3, which target only the correlation computation step.

Fuzzy FC Matrix Impact Analysis

Metric	Value
Mean absolute difference	1.656e-16
Median absolute difference	1.209e-10
Min absolute difference	2.647e-23
Max absolute difference	2.419e-08
Std of differences	7.787e-13
Unique absolute perturbation magnitudes	253102
Unique signed perturbation magnitudes (total)	357215
Unique signed perturbation magnitudes (positive)	177961 (49.8%)
Unique signed perturbation magnitudes (negative)	179253 (50.2%)
Total % of edges perturbated in all runs	100.0% (99235 edges)

Figure 15 - Effect of numerical perturbation on correlation coefficient computation.

As Figure 15 illustrates, the MCA perturbation of the `np.corrcoef` function introduces only minimal numerical noise into the resulting coefficients, with a mean absolute difference of 1.656×10^{-16} and a maximum of 2.419×10^{-8}. These results are consistent with the findings of Denis et al., 2018 [4], who demonstrated that Verificarlo's perturbations remain within the expected bounds of floating-point uncertainty. It is worth noting that the magnitude of these perturbations depends fundamentally on the numerical stability of the perturbed function itself. Less stable operations would be expected to amplify the perturbations more significantly.

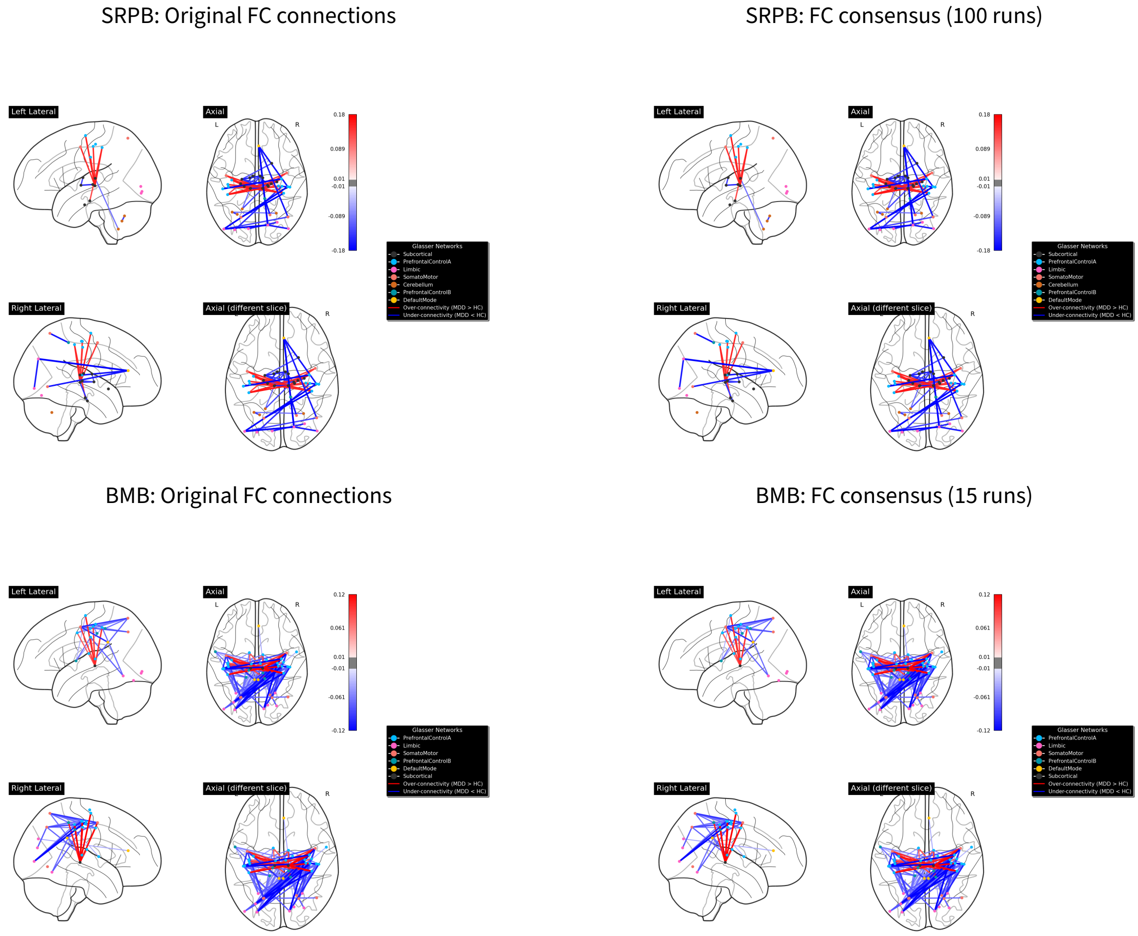


Figure 16 - Comparison of original and consensus FC connections for SRPB (top) and BMB (bottom) datasets.

Figure 16 displays the selected FC connections using the same color scheme: red for over-connectivity and blue for under-connectivity in MDD subjects relative to healthy controls. It compares the original results with the consensus connections retained across all perturbed runs for both the SRPB (100 runs) and BMB (15 runs) datasets. The original and consensus plots are identical, demonstrating that the feature extraction process is entirely robust to the introduced numerical perturbations. This is further supported by examining not only the visual representation of the FC connections, but also the computed network statistics. Both the edges per network (Figure 17) and the inter-connections (Figure 18) remain identical between the original and consensus results.

SRPB: Original edges per network

FC Edges per Network

Network	Total Edges	Intra-Network	% of Total
Subcortical	62	18	52.5%
PrefrontalControlA	28	2	23.7%
Limbic	10	3	8.5%
SomatoMotor	8	0	6.8%
Cerebellum	6	2	5.1%
DefaultMode	3	0	2.5%
PrefrontalControlB	1	0	0.8%

SRPB: Edges per network consensus (100 runs)

FC Edges per Network

Network	Total Edges	Intra-Network	% of Total
Subcortical	62	18	52.5%
PrefrontalControlA	28	2	23.7%
Limbic	10	3	8.5%
SomatoMotor	8	0	6.8%
Cerebellum	6	2	5.1%
DefaultMode	3	0	2.5%
PrefrontalControlB	1	0	0.8%

BMB: Original edges per network

FC Edges per Network

Network	Total Edges	Intra-Network	% of Total
PrefrontalControlA	128	11	42.1%
SomatoMotor	66	8	21.7%
PrefrontalControlB	38	3	12.5%
Limbic	33	0	10.9%
Subcortical	28	0	9.2%
DefaultMode	11	0	3.6%

BMB: Edges per network consensus (15 runs)

FC Edges per Network

Network	Total Edges	Intra-Network	% of Total
PrefrontalControlA	128	11	42.1%
SomatoMotor	66	8	21.7%
PrefrontalControlB	38	3	12.5%
Limbic	33	0	10.9%
Subcortical	28	0	9.2%
DefaultMode	11	0	3.6%

Figure 17 - Comparison of original and consensus edges per network for SRPB (top) and BMB (bottom) datasets.

SRPB: Original inter-connections

Top Inter-Network Connections

Connection Pair	Count
PrefrontalControlA ↔ Subcortical	20
SomatoMotor ↔ Subcortical	4
Limbic ↔ PrefrontalControlA	2
Cerebellum ↔ Subcortical	2
PrefrontalControlB ↔ SomatoMotor	1
DefaultMode ↔ Limbic	1
DefaultMode ↔ PrefrontalControlA	1
DefaultMode ↔ SomatoMotor	1
Limbic ↔ SomatoMotor	1
PrefrontalControlA ↔ SomatoMotor	1

SRPB: Inter-connections consensus (100 runs)

Top Inter-Network Connections

Connection Pair	Count
PrefrontalControlA ↔ Subcortical	20
SomatoMotor ↔ Subcortical	4
Limbic ↔ PrefrontalControlA	2
Cerebellum ↔ Subcortical	2
PrefrontalControlB ↔ SomatoMotor	1
DefaultMode ↔ Limbic	1
DefaultMode ↔ PrefrontalControlA	1
DefaultMode ↔ SomatoMotor	1
Limbic ↔ SomatoMotor	1
PrefrontalControlA ↔ SomatoMotor	1

BMB: Original inter-connections

Top Inter-Network Connections

Connection Pair	Count
PrefrontalControlA ↔ SomatoMotor	32
Limbic ↔ PrefrontalControlA	26
PrefrontalControlA ↔ Subcortical	22
PrefrontalControlA ↔ PrefrontalControlB	19
PrefrontalControlB ↔ SomatoMotor	8
DefaultMode ↔ PrefrontalControlA	7
SomatoMotor ↔ Subcortical	4
DefaultMode ↔ SomatoMotor	3
Limbic ↔ SomatoMotor	3
Limbic ↔ PrefrontalControlB	3

BMB: Inter-connections consensus (15 runs)

Top Inter-Network Connections

Connection Pair	Count
PrefrontalControlA ↔ SomatoMotor	32
Limbic ↔ PrefrontalControlA	26
PrefrontalControlA ↔ Subcortical	22
PrefrontalControlA ↔ PrefrontalControlB	19
PrefrontalControlB ↔ SomatoMotor	8
DefaultMode ↔ PrefrontalControlA	7
SomatoMotor ↔ Subcortical	4
DefaultMode ↔ SomatoMotor	3
Limbic ↔ SomatoMotor	3
Limbic ↔ PrefrontalControlB	3

Figure 18 - Comparison of original and consensus inter-connections for SRPB (top) and BMB (bottom) datasets.

Overall, the reproduction of the PCA-based feature extraction on the complete SRPB and BMB datasets confirmed the general pattern of under-connectivity in MDD subjects observed in the original study, while revealing a broader and more variable involvement of prefrontal, motor, and subcortical networks. Crucially, the perturbation analysis demonstrated that this pipeline is highly numerically stable. The combined numerical noise introduced during both the correlation computation and the 32-bit PCA dimensionality reduction did not propagate to the final selected biomarkers, which remained entirely unchanged across all perturbed runs. This robustness suggests that the method is reliable and resilient to the floating-point variations typically encountered across different computational environments.

Chapter 5

Conclusion

This bachelor thesis successfully achieved its three main objectives. First, it reproduced the findings of Alizadeh et al., 2026 [1], identifying and correcting a bug in the original NPVR simulation pipeline. The stability of graph metrics was further evaluated using a different multi-site dataset and a more extensive confound regression strategy. Despite these methodological changes, the analyzed graph metrics exhibited trends consistent with the original study, while notably demonstrating that applying a larger number of confounds significantly increases the overall NPVR and amplifies its variability across thresholds. Second, it extended the stability analysis to the FC matrices themselves, revealing a substantial numerical variability of approximately 11% of the population variability across individual edges when using all 15 confounds. While the graph metrics assessment suggested that adding confounds generally increases numerical variability, the global-signal regression proved to be a notable exception, as its inclusion effectively halved the NPVR across the vast majority of edges. Third, it evaluated the numerical stability of the PCA feature extraction of MDD biomarkers proposed by Yamashita et al., 2026 [2] by perturbing two distinct steps of its pipeline, as well as its generalization when applied to the complete SRPB and BMB datasets. This method consistently selected FC connections across similar but broader brain networks, characterized by widespread under-connectivity in MDD subjects. However, the relative contribution of these networks varied between datasets. In the SRPB dataset, the Subcortical network was the most represented, with its connection to the PrefrontalControlA network being the most frequent. In the BMB dataset, the PrefrontalControlA network itself, along with its connections to the SomatoMotor network, emerged as the most significant. The perturbations introduced during the correlation computation and the 32-bit PCA dimensionality reduction had no effect on the final extracted biomarkers. This confirms that while the specific connectivity patterns vary across datasets, the pipeline's results remain entirely robust to numerical perturbations.

Building on these findings, several promising directions for future research emerge. First, while this study highlighted the contrasting effects of global signal regression, a systematic assessment of the individual contributions of each confound to the NPVR would be highly valuable. Understanding which specific regressors introduce the most numerical noise could guide the optimization of preprocessing pipelines for maximum stability. Second, the numerical robustness of the PCA feature extraction pipeline could be further stress-tested. Beyond the correlation computation and precision reduction evaluated here, future work could introduce perturbations into the cross-site harmonization step, as well as the confound regression process itself. Ultimately, these extensions would further solidify the reliability of these biomarkers for clinical applications.

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Appendix

Table of Acronyms

ACP :	Analyse en Composantes Principales
ATR :	Advanced Telecommunications Research Institute International
CF :	Connectivité Fonctionnelle
FC :	Functional Connectivity
FD :	Framewise Displacement
FDR-BH :	False Discovery Rate with Benjamini-Hochberg
FWHM :	Full Width at Half Maximum
HPC :	High Performance Computing
IRM :	Imagerie par Résonance Magnétique
IRMf :	Imagerie par Résonance Magnétique Fonctionnelle
KKT :	Karush–Kuhn–Tucker
LLM :	Large Language Model
MCA :	Monte Carlo Arithmetic
MDD :	Major Depressive Disorder
MRI :	Magnetic Resonance Imaging
NPVR :	Numerical Population Variability Ratio
OS :	Operating System
PC :	Principal Component
PCA :	Principal Component Analysis
PR :	Pull Request
ROI :	Region Of Interest
SE :	Système d'Exploitation
TDM :	Trouble Dépressif Majeur
fMRI :	Functional Magnetic Resonance Imaging

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